Choose a motion method

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose a motion relationship and justify its assumptions.

SUCCESS CRITERIA - I can...

1. Choose a relationship using the known quantities.
2. Show linked calculations with units and directions.
3. State when constant velocity or acceleration applies.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “constant” and “interval” to explain your model or result. Sentence starters are optional.

constant: not changing during the stated interval

interval: the time or range between two stated values

acceleration: rate of change of velocity

01 Warm-up: bring back an earlier idea

A cart starts at 3 m and moves right at 5 m/s for 2 s. **Predict** final position. Allowed endpoints are 11 to 15 m; state this interval and one assumption.

Word help: final = at the end of the interval; subscript f

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: ① ② ③

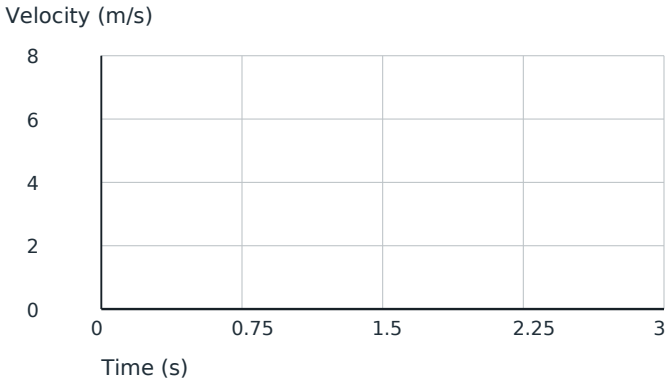
02 Worked example: explain the model

Reference: $v_f = v_i + a\,\Delta t$; $\Delta x = \frac{1}{2}(v_i + v_f)\Delta t$. Constant acceleration; i = initial, f = final. At constant velocity, $\Delta x = v\,\Delta t$.

Worked reasoning: From rest, a cart accelerates uniformly at 2 m/s² for 3 s. Final velocity=6 m/s; average velocity=(0+6)/2=3 m/s; displacement=3×3=9 m.

Your turn: Complete the cart's velocity-time graph. Shade the triangular area under the line and use it to check the 9 m displacement.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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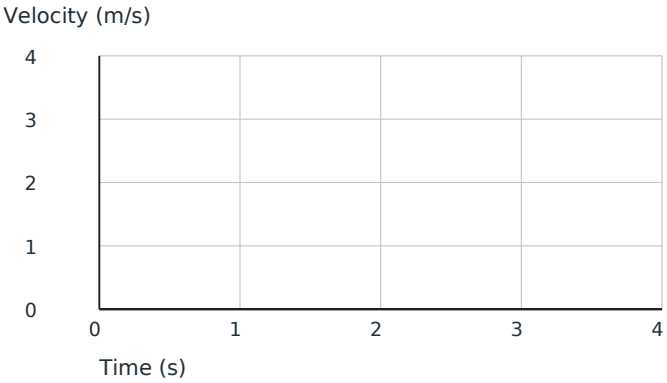
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Choose the relationship

A robot drives straight at a steady 3 m/s for 4 s.
Sketch its velocity-time graph. Decide whether a constant-velocity or acceleration relationship fits, then find its displacement.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

04 Build final velocity

A remote-control car starts at 2 m/s east and gains 2 m/s each second for 3 s. Complete the table. How much velocity does it gain altogether, and what is its final velocity?

Sentence starter: I notice … . This supports … because … .

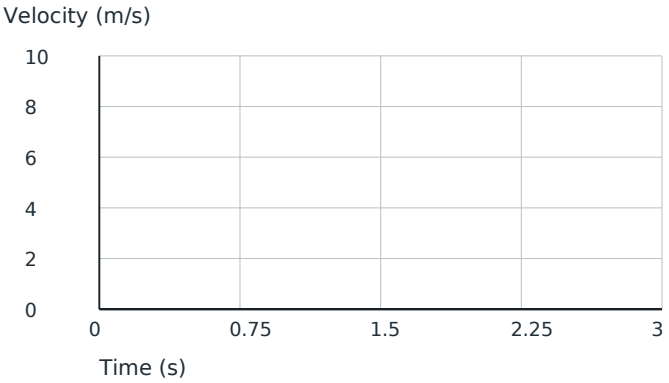
Time (s)	Velocity east (m/s)
0	2
1	4
2	6
3	?

Marzano check - circle: ① ② ③

05 Find the distance while speeding up

Use the car's velocities from question 4 to complete the graph. Its acceleration is constant.
Find its average velocity and displacement over the 3 s.
Use the graph's area to check.

Sentence starter: I found … [unit]. This value means … in this situation.

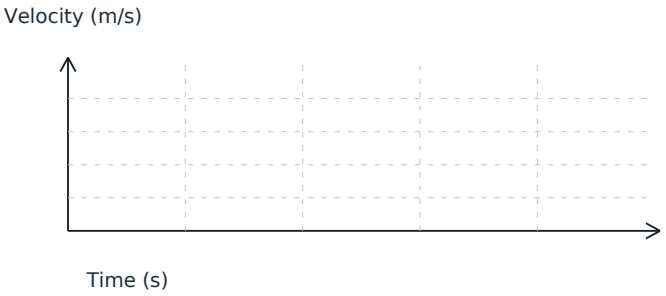


Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A cart travels west at 10 m/s and west at 3 m/s 2 s later. Is it speeding up?
Find acceleration size and direction.
Sketch the velocity-time model; label the given values.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

Choose a motion method

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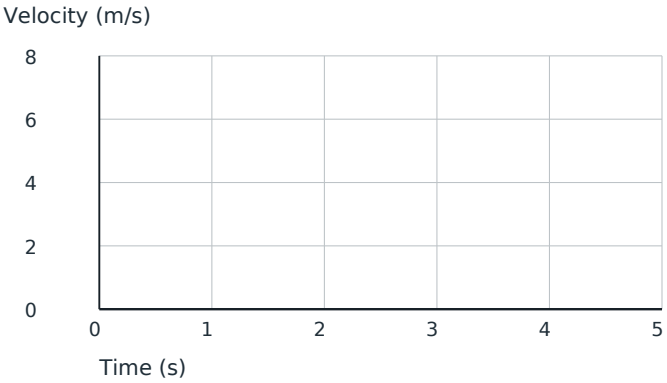
Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

In a launch test, a small rover speeds up steadily from rest to 6 m/s east in 5 s. Plot the two endpoints and connect them.
Find acceleration, average velocity and displacement.

Sentence starter: I found … [unit]. This value means … in this situation.

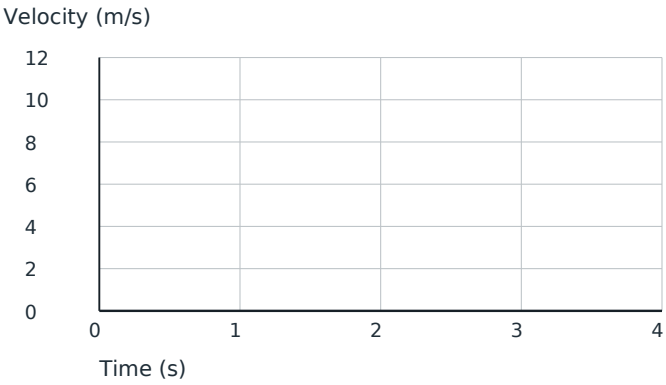


Marzano check - circle: 1 2 3

08 STRETCH: test the model

Create two different motions that travel 8 m in 4 s: one at steady speed and one speeding up uniformly from rest.
Draw their velocity-time graphs.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3